

	ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)		LESSON PLAN
Programme	B.Tech.	Academic Year	2025-26
Department	CSE/CSIT	Semester	6th
Credit	4	Grading Pattern	1
Subject Code	CSE 3151		
Subject Name	Introduction To Databases		
Weekly Course Format	3L - 1P		
Subject Coordinator (s)	Mr. Rakesh Kumar		

Text Books(s):

- (1) Database System Concepts, 6th Edition, by Silberschatz, Korth, Sudharsan, Tata Mc-Graw Hills Publication.
- (2) Learning SQL: Master SQL Fundamentals, 2nd Edition, by Alan Beaulieu, O Reilly.

Course Outcomes	Students will be able to	
	CO1	Identify and explain the different components and functionalities of DBMS and their interdependence through the database architecture.
	CO2	Model SQL queries and able to develop a database application as per user requirements using SQL and database connectivity.
	CO3	Analyze an enterprise schema for given user requirements and apply the conceptual database design principles through ER modelling to construct the ER diagram and corresponding relation schemas.
	CO4	Analyze and design relational database schema using decomposition and normalization techniques.
	CO5	Apply relational algebra and relational calculus to express queries on relational schemas.
	CO6	Interpret the functional issues related to transaction and database recovery along with the concept of storage and database system architecture.

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Sl.No.	Lessons/Topics to be covered	Book Reference (sections)	Mapping with COs	Home Work/ Assignments/ Quizzes
1	Introduction to the course and its motivation. Departmental Vision and Mission, course description, objective, course outcomes, credit, grading pattern, class and Lab session of the course. NBA provided program outcomes and departmental specific program specific outcomes (PSO), Program educational objectives (PEO).		All Cos, POs, PSOs	
2	Introduction to DBMS:- Data, Database, Database system, Application, Disadvantages of conventional FileSystem, Advantages of Database approach on File system.	(Chapter1), Silber-schatz, Korth & Sudharsan (pg.1-5)	CO1	
3	Three level Data abstraction, Schema, Instances, Data Independence	(Chapter1), Silber-schatz, Korth & Sudharsan (pg.6-8)	CO1	
4	Lab#1: Assignment 1: Practicing DDL and DML Commands used for table creation and insertion	(Chapter3), Silber-schatz, Korth & Sudharsan, (Chapter2), Alan Beaulieu	CO2	
5	Database Languages: DDL, DML, Database users, DBA,	(Chapter1), Silber-schatz, Korth & Sudharsan (pg.9-15)	CO1	

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6	Data Models: ER model, Relational Model, Network model, hierarchical model, Object based data model, semi structured data model, Relational databases, Database Design	(Chapter1), Silber-schatz, Korth & Sudharsan (pg.9)	CO1	
7	Overall DB system architecture, Data Storage and querying Two tier and three tier architecture	(Chapter1), Silber-schatz, Korth & Sudharsan (pg.24-25)	CO1	
8	Lab#2: Assignment 2: Practicing DDL and DML commands for writing queries	(Chapter3), Silber-schatz, Korth & Sudharsan, (Chapter2), Alan Beaulieu	CO2	
9	ER Model:- Entity, attribute, Entity set, Relationship, Relationship set, degree of relationship set, types of attributes with examples	(Chapter7), Silber-schatz, Korth & Sudharsan (pg.262-268)	CO3	
10	Mapping cardinalities Participation constraint, Database Keys: - Super key, Candidate key, Primary key, Alternate Key, Secondary key	(Chapter 2, 7), Silberschatz, Korth & Sudharsan (pg.269-278)	CO3	QUIZ 1

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11	Foreign key, Referential integrity, Key attributes of relationship set, Strong, Weak entity, Identifying relationship	(Chapter7), Silber-schatz, Korth & Sudharsan (pg.279-281)	CO3	
12	Lab#3: Assignment 2 contd.	(Chapter3), Silber-schatz, Korth & Sudharsan, (Chapter2), Alan Beaulieu	CO2	
13	EER Features:- Specialization, Generalization, ISA relationship , Constraints on Specialization, Generalization, Aggregation	(Chapter7), Silber-schatz, Korth & Sudharsan (pg.295-302)	CO3	
14	Designing an ER diagram using EER and ER features as a case study	(Chapter7), Silber-schatz, Korth & Sudharsan (pg.282)	CO3	Assignment-1
15	Relational database:- What is a Relation, Different features of a Relation(Attribute, Domain, Null value, Tuple), Relational Schema , Relational Database Constraints	(Chapter 2, 7), Silber-schatz, Korth & Sudharsan (pg.39-42)	CO4	

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Sl.No.	Lessons/Topics to be covered	Book Reference (sections)	Mapping with COs	Home Work/ Assignments/ Quizzes
16	Lab#4: Assignment 3: Practicing SQL queries using SQL operators, Aggregate and Scalar functions	(Chapter3), Silber-schatz, Korth & Sudharsan, (Chapter6), Alan Beaulieu	CO2	
17	Mapping ER model to Relational Model with examples and case study	(Chapter7), Silber-schatz, Korth & Sudharsan (pg.283-288)	CO4	QUIZ-2
18	Relational Database Design:- Anomalies in Database(Insertion, Deletion, Updation Anomalies with examples), Normalization concept, Decomposition Method,	(Chapter8), Silber-schatz, Korth & Sudharsan (pg.323-325)	CO4	
19	Functional Dependencies, Trivial functional dependency, Inference Rules for FDs, Closure of FD	(Chapter8), Silber-schatz, Korth & Sudharsan (pg.325-340)	CO4	

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Sl.No.	Lessons/Topics to be covered	Book Reference (sections)	Mapping with COs	Home Work/ Assignments/ Quizzes
20	Lab#5: Assignment 3 contd.	(Chapter3), Silber-schatz, Korth & Sudharsan, (Chapter7), Alan Beaulieu	CO2	
21	Attribute closure, Extraneous attribute, Canonical Cover	(Chapter8), Silber-schatz, Korth & Sudharsan (pg.340-345)	CO4	
22	Decomposition, Properties of Decomposition, Loss less decomposition, Dependency preserving decomposition	(Chapter8), Silber-schatz, Korth & Sudharsan (pg.345-348)	CO4	
23	Normalization, INF, Second normal form	(Chapter8), Silber-schatz, Korth & Sudharsan (pg.328-334)	CO4	

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Sl.No.	Lessons/Topics to be covered	Book Reference (sections)	Mapping with COs	Home Work/ Assignments/ Quizzes
24	Lab#6: Assignment 3 contd.	(Chapter3), Silber-schatz, Korth & Sudharsan, (Chapter8), Alan Beaulieu	CO2	
25	3NF, BCNF, comparison of 3NF with BCNF	(Chapter8), Silber-schatz, Korth & Sudharsan (pg.333-349,354)	CO4	
26	Multivalued dependency, inference rules to find closure of multivalued dependency set	(Chapter8), Silber-schatz, Korth & Sudharsan (pg.355-358)	CO4	
27	4NF, Problems based on normalization, Denormalization	(Chapter8), Silber-schatz, Korth & Sudharsan (pg.358-360)	CO4	QUIZ-3

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Sl.No.	Lessons/Topics to be covered	Book Reference (sections)	Mapping with COs	Home Work/ Assignments/ Quizzes
28	Lab#7: Assignment 4: Table creation using Database Constraints	(Chapter4), Silber-schatz, Korth & Sudharsan, (Chapter13), Alan Beaulieu	CO2	
29	Relational Query Language:- Procedural and Non- Procedural query language, Relational Algebra :- Selection, Projection, union	(Chapter6), Silber-schatz, Korth & Sudharsan (pg.217-228)	CO5	
30	Relational algebra queries based on Set difference, Cartesian product, Rename	(Chapter6), Silber-schatz, Korth & Sudharsan (pg.228-236)	CO2	
31	Relational algebra queries based on Set intersection, Assignment, Natural Join, Extended relational algebra operations	(Chapter6), Silber-schatz, Korth & Sudharsan (pg.236-239)	CO2	

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Sl.No.	Lessons/Topics to be covered	Book Reference (sections)	Mapping with COs	Home Work/ Assignments/ Quizzes
32	Lab#8: Assignment 4 contd.	(Chapter4), Silber-schatz, Korth & Sudharsan, (Chapter13), Alan Beaulieu	CO2	
33	Queries using Tuple relational calculus, Natural Join in TRC	(Chapter6), Silber-schatz, Korth & Sudharsan (pg.239-244)	CO5	
34	Domain Relational Calculus, Natural Join in Domain relational calculus	(Chapter6), Silber-schatz, Korth & Sudharsan (pg.245-248)	CO5	
35	Transaction, ACID Properties of transaction, States of Transaction	(Chapter14), Silber-schatz, Korth & Sudharsan (pg.627-635)	CO6	

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Sl.No.	Lessons/Topics to be covered	Book Reference (sections)	Mapping with COs	Home Work/ Assignments/ Quizzes
36	Lab#9: Assignment 5: Practicing DML commands using sub queries and Join	(Chapter4), Silber-schatz, Korth & Sudharsan, (Chapter4,9), Alan Beaulieu	CO2	
37	Transaction Schedule, serial and concurrent schedule, Serializability	(Chapter14), Silber-schatz, Korth & Sudharsan (pg.636-640)	CO6	
38	Conflict Serializability, View serializability	(Chapter14), Silber-schatz, Korth & Sudharsan (pg.641-646)	CO6	
39	Recoverability, Recoverable and Cascade less Schedule with example	(Chapter14), Silber-schatz, Korth & Sudharsan (pg.647-648)	CO6	Assignment-2

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Sl.No.	Lessons/Topics to be covered	Book Reference (sections)	Mapping with COs	Home Work/ Assignments/ Quizzes
40	Lab#10: Assignment 5 contd.	(Chapter4), Silber-schatz, Korth & Sudharsan, (Chapter5,10), Alan Beaulieu	CO2	
41	Concurrency control: Need, Lock based concurrency control, 2 phase locking protocol, strict and rigorous 2phase locking protocol	(Chapter15), Silber-schatz, Korth & Sudharsan (pg.661-668)	CO6	
42	Time-Stamp-based concurrency control, Multi version Scheme protocol, advantages and performance	(Chapter15), Silber-schatz, Korth & Sudharsan (pg.682-691)	CO6	QUIZ-4
43	Recovery System:- DB storage and failure, Log-based Recovery, REDO and UNDO operations	(Chapter16), Silber-schatz, Korth & Sudharsan (pg.721-733)	CO6	
44	Lab#11: Assignment 6: Project Based on Database Connectivity	(Chapter5), Silber-schatz, Korth & Sudharsan	CO2	

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45	Recovery in Deferred Database Modification and Immediate Database Modification, Check Pointing.	(Chapter16), Silber-schatz, Korth & Sudharsan (pg.730-734)	CO6	
46	Primary and secondary index, dense and sparse index, multilevel indexing, B+ tree index, Hashing	(Chapter11), Silber-schatz, Korth & Sudharsan (pg.475-488)	CO6	
47	Query Processing;Evaluation of query cost, Query Optimization: Overview, Transformation of Relational expressions	(Chapter12-13), Silber-schatz, Korth & Sudharsan (pg.438,579-581)	CO6	
48	Lab#12: Assignment 6 contd.	(Chapter5), Silber-schatz, Korth & Sudharsan	CO2	

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